

SMART INDIA HACKATHON 2026

R.A.I.

Rainfall Artificial Intelligence

Explainable AI-Based Rainfall Intelligence & Decision Support Platform

"R.A.I. converts weather forecasts into explainable decisions — not just 'Will it rain?', but 'Could it become dangerous, why, where, and what should we do?'"

 **168h**
Telemetry

 **TreeSHAP**
Explainable AI

 **Agronomic**
Farmer AI

IN Dual
Hindi / English

SIH 2026 SUBMISSION DETAILS

PROBLEM STATEMENT ID

[SIH-2026 Problem Statement ID]

PROBLEM STATEMENT TITLE

AI-Powered Climate Intelligence & Precision Extreme Weather Decision Support Platform

THEME

Agriculture, Climate Tech & Disaster Management

PS CATEGORY

Software Edition

TEAM ID

[Your Team ID]

TEAM NAME

[Your Team Name]

PROPOSED SOLUTION: EXPLAINABLE RAINFALL INTELLIGENCE

Transforming atmospheric data into transparent, calibrated risk intelligence and automated decision guidance.

1. Live Telemetry

168h Open-Meteo feeds
Pressure, wind, clouds

2. Heavy-Rain ML

Calibrated XGBoost
Target: ≥ 64.5 mm/day

3. Explainable AI

TreeSHAP attribution
Feature driver weights

4. Spatial Risk Map

Catchment hydrology
Elevation flood contours

5. Decision Support

Farmer agronomic AI
Emergency action routes

01. Intelligence

Weather & XAI

Multi-horizon precipitation forecasting engine with TreeSHAP driver explanations and confidence intervals.

02. Risk Map

Geospatial Hydrology

Catchment basin run-off modeling, elevation flood vectors, and real-time storm hotspot visualization.

03. Emergency

Life Safety Broadcast

Civil defense-aligned warning protocols directing evacuations, safe-haven capacities, and hazard havens.

04. Relief

Civic Logistics

Essential resource shortage tracking (sandbags, clean water) synchronizing NGO aid with municipal logistics.

05. Farmer AI

Agronomic Guidance

Soil saturation forecasts, crop vulnerability analysis, and precision field drainage guidance.

CRITICAL SCIENTIFIC & CONCEPTUAL DISTINCTION IN R.A.I.

- Ordinary Precipitation Probability (e.g. 57%–75%): Chance of any measurable rain (≥ 0.1 mm) in the forecast window.
 - Extreme Heavy-Rain ML Risk (e.g. 0.9% vs 1.5% Threshold): Dedicated calibrated XGBoost detecting dangerous deluge (≥ 64.5 mm/day).
- ⇒ R.A.I. never conflates these two metrics — ensuring clear, accurate risk comprehension without false public panic.

TECHNICAL APPROACH: ML PIPELINE, XAI & KNOWLEDGE RETRIEVAL

1. CALIBRATED ML ENGINE

- **Production Model:** RAI-HeavyRain-XGBoost-IMD (v1.0.0-sih-xgb)
- **14-Feature Vector:** Surface pressure, wind vectors, dewpoint, 3-layer cloud cover, humidity, historical delta
- **Heavy-Rain Target:** ≥ 64.5 mm/day (IMD Extreme Benchmark)
- **Operational Threshold:** 0.0150 (1.5% calibrated threshold)
- **Calibration Method:** Isotonic Regression (Zadrozny & Elkan)
- **IMD Rainfall Tiers:** Light (0.1–7.5), Moderate (7.6–64.4), Heavy (64.5–115.5), Very Heavy (115.6–204.4), Extreme (≥ 204.5 mm)

2. EXPLAINABLE AI (XAI)

- **Explainability Engine:** TreeSHAP Additive Attributions (Lundberg & Lee)
- **Attribution Output:** Exact numerical SHAP values per atmospheric feature (+/- impact on log-odds)
- **Dual Factor Breakdown:** Increasing factors (pressure drop, moisture surge) vs Reducing factors (dry wind)
- **Scientific Safety Rule:** TreeSHAP quantifies statistical association with model predictions — it does NOT claim physical causation.
- **Natural Interpretation:** Translates raw feature math into plain Hindi & English natural language summaries.

3. KNOWLEDGE & DIALOGUE

- **Knowledge Base:** 35,857 structured meteorological & agronomic domain records
- **Retrieval Architecture:** Token inverted indexing, normalized routing, TF-IDF/BM25 weighting & category boost
- **Conversational AI:** Multi-turn context memory, typo/Hinglish tolerance, proactive follow-up suggestions
- **Telemetry Injection:** Live Open-Meteo values automatically embedded into knowledge responses
- **Bilingual Parity:** Full English & natural conversational Hindi support across all 5 pillars

END-TO-END DATAFLOW PIPELINE ARCHITECTURE

Live Telemetry → 14-Feature Production Transform → XGBoost Probability → Isotonic Calibration (1.5% Threshold) → TreeSHAP Attributions → Spatial Hydro-Mapping → Farmer/Emergency Advisory

FEASIBILITY & VIABILITY: IMPLEMENTED • TESTED • VERIFIABLE

R.A.I. is backed by rigorous automated verification suites, modular architecture, and honest engineering disclosures.

1,262 / 1,262

System Assertions Passed

1,128 / 1,128

Knowledge Retrieval Tests

35 / 35

Secure Auth & OTP Tests

34 / 34

Bilingual Parity Tests

1,754 Modules

Compiled (0 Errors)

TECHNICAL FEASIBILITY

- ✓ **Zero Hardware Deployment:** Runs on any modern device using coordinate-based live Open-Meteo telemetry.
- ✓ **Modular Architecture:** Clean separation of ML inference, spatial rendering, conversational routing, and auth layers.
- ✓ **Sub-Second Latency:** Local in-memory indexing delivers instantaneous knowledge responses (<30ms).
- ✓ **Salted Password Hashing:** PBKDF2/Argon2-ready server derivation with unique 128-bit cryptographic salts.

ENGINEERING DISCLOSURES

- ⚠ **Development OTP Simulator:** Production delivery requires real SMTP/SMS gateway credentials (e.g. Twilio/SendGrid).
- ⚠ **Specialized Heavy-Rain ML:** Trained specifically for extreme events (≥ 64.5 mm); does not pretend to forecast general synoptics.
- ⚠ **Attribution vs Causation:** TreeSHAP calculates statistical feature contributions rather than proving physical meteorology.
- ⚠ **Open Telemetry Dependent:** Requires active internet access to synchronize live Open-Meteo forecasts.

FUTURE SCALING ROADMAP

- 🚀 **Multi-Hazard Expansion:** Extending ML models to Agricultural Drought, Heatwaves, and Dry-Spell Duration.
- 🚀 **IoT Ground Sensors:** Integrating low-cost LoRaWAN soil moisture probes for hyper-local microclimate calibration.
- 🚀 **NDMA & State SDMA APIs:** Direct automated webhook alerts to State Disaster Management Authorities.
- 🚀 **Multi-Lingual India:** Scaling beyond Hindi/English to Tamil, Telugu, Marathi, Bengali, and Punjabi.

IMPACT & BENEFITS: CONVERTING INTELLIGENCE INTO DECISIONS

FARMERS & GROWERS

- **Irrigation Timing:** Hold irrigation when heavy rain is forecast; saves water and power.
- **Drainage Protection:** Clear field channels before deluge to prevent root rot & crop lodging.
- **Fertilizer Efficiency:** Avoid applying nitrogen before downpours, preventing nutrient runoff.

EVERYDAY CITIZENS

- **Hourly Precision:** Understand rain onset windows for daily commuting and outdoor safety.
- **Flood Route Safety:** Avoid inundation hotspots using live spatial hydrological vectors.
- **Explainable Alerts:** Clear plain-language explanations instead of ambiguous weather jargon.

EMERGENCY TEAMS

- **Tiered Warning Protocols:** Advisory → Watch → Warning → Evacuation escalation stages.
- **Safe Haven Capacities:** Verified shelter locations with live capacity and routing.
- **Civil Defense Sync:** Rapid resource deployment matching municipal response needs.

MUNICIPAL & RELIEF

- **Catchment Modeling:** Predict dynamic run-off vectors before storm drains overflow.
- **Shortage Tracking:** Real-time auditing of essential relief supplies (water, sandbags).
- **NGO Coordination:** Verified distribution hubs to eliminate duplicate aid efforts.

FARMER AI ESTABLISHED DECISION FRAMEWORK & CORE DIFFERENTIATION

- ✓ DO THIS (Recommended): Hold irrigation if heavy precipitation is predicted; inspect and unblock field drainage channels.
- ✗ AVOID THIS (High Risk): Do not apply water-soluble nitrogen fertilizers immediately before intense rain (prevents leaching).
- CHECK LATER (Re-assess): Re-evaluate soil saturation 24h post-rainfall before restarting scheduled irrigation cycles.

“Ordinary weather apps state whether rain is likely. R.A.I. quantifies extreme-rainfall risk, explains contributing atmospheric drivers via XAI, maps spatial flood vectors, and translates intelligence into clear, life-saving decisions.”

RESEARCH FOUNDATIONS & SCIENTIFIC REFERENCES

R.A.I. is grounded in peer-reviewed meteorological, machine learning, and explainable AI literature.

1. METEOROLOGY & TELEMETRY STANDARDS

[1] India Meteorological Department (IMD): Rainfall intensity classification criteria & extreme thresholds (Light: 0.1-7.5, Mod: 7.6-64.4, Heavy: 64.5-115.5, Very Heavy: 115.6-204.4, Extreme: ≥ 204.5 mm/day).

[2] Open-Meteo Weather API: Global Atmospheric Telemetry Documentation (CC BY 4.0), integrating ECMWF Integrated Forecasting System (IFS) and GFS numerical models.

2. MACHINE LEARNING & PROBABILITY CALIBRATION

[1] Chen, T., & Guestrin, C. (2016): XGBoost: A Scalable Tree Boosting System. In Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD).

[2] Zadrozny, B., & Elkan, C. (2002): Transforming classifier scores into accurate multiclass probability estimates (Isotonic Calibration). ACM SIGKDD.

3. EXPLAINABLE ARTIFICIAL INTELLIGENCE (XAI)

[1] Lundberg, S. M., & Lee, S.-I. (2017): A Unified Approach to Interpreting Model Predictions (TreeSHAP). Advances in Neural Information Processing Systems (NeurIPS 30).

[2] Additive Feature Attribution: Game-theoretic Shapley value attribution for tree ensembles, isolating individual atmospheric feature contributions to log-odds predictions.

4. AGRONOMIC & DISASTER DECISION SUPPORT

[1] Smart India Hackathon (SIH 2026): Government of India AI-Powered Disaster Management, Climate Resilience and Precision Agriculture Challenge Framework.

[2] ICAR Agronomic Literature: Soil moisture saturation thresholds, root lodging dynamics, and crop phenological vulnerability stages under acute precipitation events.